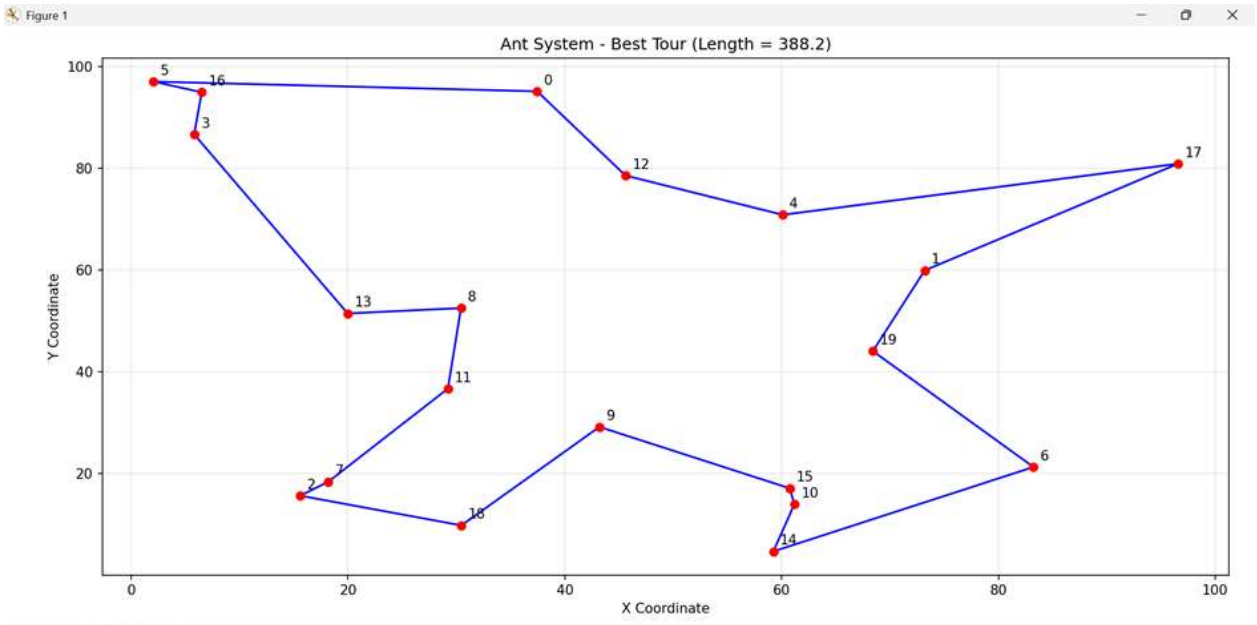
Ant Colony Optimization for the Traveling Salesman Problem




```
PS C:\Users\DELL\OneDrive\Desktop\BIS LAB> python -u "c:\Users\DELL\OneDrive\Desktop\BIS LAB\Particle Swarm Optimization for Function Optimization\PSO.py"

========== PSO RESULTS ==========
Optimal x: -0.000000073064582
Optimal y: 0.000000140758272
Minimum fitness: 2.5151324137874 56e-14

======= FITNESS PER ITERATION =======
Iteration               Best Fitness
1               1.004062079380507e+00
2               1.933440031348827e-01
3               8.203772388816193e-02
4               7.309505249386285e-02
5               7.309505249386285e-02
6               2.573245250071099e-02
7               1.521667678394887e-02
8               2.930195712056611e-03
9               2.930195712056611e-03
10              2.930195712056611e-03
11              2.930195712056611e-03
12              2.930195712056611e-03
13              2.930195712056611e-03
14              2.930195712056611e-03
15              2.930195712056611e-03
16              2.930195712056611e-03
17              2.930195712056611e-03
18              6.005675986779610e-04
19              6.005675986779610e-04
20              6.005675986779610e-04
21              1.911288619883584e-04
22              4.342136058176498e-05
23              4.342136058176498e-05
24              4.342136058176498e-05
25              4.342136058176498e-05
26              2.026092109208630e-05
27              2.026092109208630e-05
28              2.026092109208630e-05
29              1.658343352947658e-06
30              1.658343352947658e-06
31              1.658343352947658e-06
32              1.658343352947658e-06
33              1.658343352947658e-06
34              1.576153519874727e-06
35              1.378641116992507e-06
36              1.065457078269369e-06
37              1.065457078269369e-06
38              8.130031325966502e-07
39              6.171099217453482e-08
40              6.171099217453482e-08
41              2.543232183121422e-08
42              1.469761049737312e-09
43              1.469761049737312e-09
44              1.469761049737312e-09
45              1.469761049737312e-09
```

```
PS C:\Users\DELL\OneDrive\Desktop\BIS LAB> python -u "c:\Users\DELL\OneDrive\Desktop\BIS LAB\Particle Swarm Optimization for Function Optimization\PSO.py"
51              1.469761049737312e-09
52              1.469761049737312e-09
53              1.469761049737312e-09
54              1.469761049737312e-09
55              1.469761049737312e-09
56              1.469761049737312e-09
57              1.469761049737312e-09
58              2.024604816190056e-10
59              2.024604816190056e-10
60              2.024604816190056e-10
61              2.024604816190056e-10
62              2.024604816190056e-10
63              2.024604816190056e-10
64              2.024604816190056e-10
65              2.024604816190056e-10
66              8.491799381060809e-12
67              8.491799381060809e-12
68              8.491799381060809e-12
69              8.491799381060809e-12
70              8.491799381060809e-12
71              8.491799381060809e-12
72              8.491799381060809e-12
73              8.491799381060809e-12
74              8.491799381060809e-12
75              8.491799381060809e-12
76              8.491799381060809e-12
77              6.647270015721526e-12
78              6.647270015721526e-12
79              6.647270015721526e-12
80              6.647270015721526e-12
81              5.522754286655824e-12
82              5.522754286655824e-12
83              5.522754286655824e-12
84              1.253874948729202e-12
85              1.253874948729202e-12
86              1.253874948729202e-12
87              1.253874948729202e-12
88              1.253874948729202e-12
89              1.253874948729202e-12
90              3.198188497737166e-14
91              3.198188497737166e-14
92              3.198188497737166e-14
93              3.198188497737166e-14
94              3.198188497737166e-14
95              3.198188497737166e-14
96              3.198188497737166e-14
97              2.515132413787456e-14
98              2.515132413787456e-14
99              2.515132413787456e-14
100             2.515132413787456e-14
PS C:\Users\DELL\OneDrive\Desktop\BIS LAB> python -u "c:\Users\DELL\OneDrive\Desktop\BIS LAB\Ant Colony Optimization for the Traveling Salesman Problem\ACO.py"
Best Tour: 4 → 12 → 0 → 5 → 16 → 3 → 13 → 8 → 11 → 7 → 2 → 18 → 9 → 15 → 10 → 14 → 6 → 19 → 1 → 17 → 4
Best Tour Length: 388.2 units
```